

LOGOVOY, Kazakhstan

WMO: 383430

Lat: 42.950N

Lon: 72.750E

Elev: 683

StdP: 93.38

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-17.8	-23.2	0.5	-21.0	-20.3	0.7	-17.7	5.4	-2.8	4.5	-2.7	0.3	0	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	35.5	18.4	34.0	17.8	32.6	17.3	19.4	31.7	18.7	31.1	18.1	30.3	1.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.8	23.6	14.3	11.0	23.2	13.4	10.4	22.8	58.2	31.5	55.9	31.5	53.9	30.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	4.5	3.8	-23.3	38.3	4.6	1.4	-26.6	39.2	-29.3	40.0	-31.9	40.8	-35.3	41.8		
(5)			-23.5	21.1	4.5	1.5	-26.8	22.2	-29.4	23.1	-32.0	24.0	-35.2	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-4.1	-2.0	6.2	12.1	17.1	22.2	24.5	23.1	17.7	10.8	3.7	-3.2		
	DBStd	11.16	6.23	6.53	5.37	4.49	3.56	3.14	2.70	3.08	4.14	4.81	5.79	6.32		
	HDD10.0	1599	437	337	137	29	1	0	0	0	1	47	199	411		
	HDD18.3	3309	696	571	375	190	65	6	1	2	61	237	438	669		
	CDD10.0	1871	0	0	20	92	222	366	448	407	233	72	11	0		
	CDD18.3	541	0	0	0	3	28	122	190	151	43	4	0	0		
	CDH23.3	7314	0	0	4	79	390	1554	2579	2010	618	76	4	0		
	CDH26.7	3268	0	0	1	16	93	658	1303	959	219	18	1	0		
(14) Wind	WSAvg	1.3	1.0	1.2	1.5	1.5	1.6	1.6	1.5	1.4	1.3	1.1	1.0	0.9		
(15) Precipitation	PrecAvg	382	19	25	35	57	57	38	27	17	16	33	32	26		
	PrecMax	562	49	53	66	125	119	91	78	50	45	94	82	52		
	PrecMin	209	5	7	10	11	20	8	6	2	3	1	1	2		
	PrecStd	90	11	11	13	28	24	20	18	13	12	22	21	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	15.6	23.2	28.7	31.2	36.0	37.8	36.9	33.5	29.4	22.8	12.3		
		MCWB	7.8	10.1	15.2	17.4	17.7	18.5	19.0	18.5	17.1	15.7	13.2	7.4		
	2%	DB	8.1	11.3	19.9	25.6	29.1	33.9	35.9	35.2	31.5	25.2	18.2	9.2		
		MCWB	4.6	7.1	12.4	15.7	17.0	18.1	18.7	18.0	16.2	13.9	11.2	5.4		
	5%	DB	5.6	8.6	17.5	23.1	27.3	32.2	34.3	33.6	29.3	22.5	14.7	6.6		
		MCWB	2.9	5.3	11.3	14.9	16.4	17.6	17.9	17.4	15.6	12.6	9.0	3.6		
	10%	DB	3.6	6.1	14.9	20.7	25.5	30.5	32.8	31.9	27.1	19.7	12.0	4.6		
		MCWB	1.5	3.5	10.0	13.7	16.0	17.1	17.5	16.9	14.7	11.3	7.7	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	10.1	15.2	18.1	19.3	20.1	20.7	19.7	18.0	16.2	13.8	7.6		
		MCDB	12.4	15.4	22.8	26.9	28.1	31.6	32.2	33.6	30.2	28.3	22.0	11.8		
	2%	WB	4.9	7.3	12.9	16.4	17.9	19.1	19.4	18.7	16.9	14.3	11.7	5.5		
		MCDB	7.6	10.9	19.2	24.2	27.0	30.5	32.4	32.5	29.7	24.2	17.4	9.0		
	5%	WB	3.2	5.4	11.5	15.2	17.1	18.4	18.7	18.1	15.9	13.0	9.5	3.7		
		MCDB	5.3	8.1	16.9	22.6	25.9	29.7	31.9	31.6	28.3	21.4	14.2	6.4		
	10%	WB	1.6	3.6	10.1	14.0	16.2	17.8	18.1	17.4	15.0	11.7	7.7	2.2		
		MCDB	3.5	6.1	14.7	20.2	24.6	28.8	31.0	30.4	26.1	19.1	11.6	4.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.0	10.8	11.5	13.0	13.9	14.8	15.3	15.1	13.1	10.3	9.0		
		MCDBR	11.9	12.8	15.3	16.5	15.9	16.6	17.1	17.1	17.7	17.7	15.5	12.5		
	5% WB	MCWBR	9.2	9.4	9.4	9.0	7.2	6.1	6.0	6.0	7.2	9.1	9.4	9.5		
		MCWBR	11.3	12.3	14.7	15.5	14.4	14.9	15.5	16.0	17.1	16.0	14.4	12.0		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.394	0.462	0.487	0.464	0.461	0.452	0.417	0.397	0.428	0.400	0.344		
	taud		2.098	2.014	1.907	1.899	2.031	2.072	2.106	2.192	2.191	2.016	2.028	2.112		
	Ebn at Noon		742	771	763	782	816	818	820	838	824	723	669	709		
	Edh at Noon		105	135	171	186	168	162	155	137	127	134	112	95		
(44) All-Sky Solar Radiation	RadAvg		1.99	2.90	3.93	5.05	6.32	6.94	6.99	6.52	5.16	3.22	2.00	1.62		
	RadStd		0.16	0.25	0.29	0.46	0.45	0.38	0.40	0.33	0.26	0.37	0.23	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	-0.75	-1.82	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.08	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page